

1. Use nested for loops to print a grid of emojis representing a 5x5 Instagram post feed, where each cell shows a 📷 symbol.

```
for (let i = 1; i <= 5; i++) {  
  for (let j = 1; j <= 5; j++) {  
    process.stdout.write(" 📷 ");  
  }  
  console.log();  
}
```

Output:

```
📷 📷 📷 📷 📷  
📷 📷 📷 📷 📷  
📷 📷 📷 📷 📷  
📷 📷 📷 📷 📷  
📷 📷 📷 📷 📷
```

2. Build a right-angled triangle pattern using nested loops, where each row displays increasing numbers starting from 1, similar to how a leaderboard on a gaming app shows rank numbers.

```
for (let i = 1; i <= 5; i++) {  
  for (let j = 1; j <= i; j++) {  
    process.stdout.write(j + " ");  
  }  
  console.log();  
}
```

Output:

```
1  
1 2  
1 2 3
```

1 2 3 4

1 2 3 4 5

**3. Create a pattern that prints a pyramid of stars (*) with 6 rows, centered like the loading animation you see on BookMyShow when a page is loading.

Hint: Use spaces to align the stars in the center for each row.**

```
for (let i = 1; i <= 6; i++) {  
  
    // Print spaces  
    for (let j = 1; j <= 6 - i; j++) {  
        process.stdout.write(" ");  
    }  
  
    // Print stars  
    for (let j = 1; j <= (2 * i - 1); j++) {  
        process.stdout.write("*");  
    }  
  
    console.log();  
}
```

Output:

```
  *  
  
 ***  
  
*****  
  
*****  
  
*****  
  
*****
```

4. Write code using nested loops to print a pattern of alternating 0s and 1s in a grid, like the checkered background seen in some Spotify playlist covers (e.g., for a 4x4 grid, alternate 0 and 1 in each cell).

```
for (let i = 0; i < 4; i++) {  
  for (let j = 0; j < 4; j++) {  
    process.stdout.write((i + j) % 2 + " ");  
  }  
  console.log();  
}
```

Output:

```
0 1 0 1  
1 0 1 0  
0 1 0 1  
1 0 1 0
```

5. Modify your pyramid pattern code to accept the number of rows as user input, so the user can set the height of the pyramid before printing.

```
let rows = Number(prompt("Enter number of rows:"));
```

```
for (let i = 1; i <= rows; i++) {
```

```
  // Print spaces
```

```
  for (let j = 1; j <= rows - i; j++) {
```

```
    process.stdout.write(" ");
```

```
  }
```

```
  // Print stars
```

```
  for (let j = 1; j <= (2 * i - 1); j++) {
```

```
    process.stdout.write("*");
```

```
}
```

```
console.log();
```

```
}
```

If the user enters **4**:

```
*
```

```
***
```

```
*****
```

```
*****
```

★ Easy logic to remember

For nested loops:

```
for (let i = 1; i <= rows; i++) {  
  for (let j = 1; j <= something; j++) {  
    // print  
  }  
}
```